

4.1.2 Objects

An object is a specific region of memory that can hold a fixed or variable value (or set of values). To prevent confusion, this use of the word object is different from the more general term used in object-oriented languages. Our definition of the word would encompass functions, variables, symbolic constants, user-defined data types, and labels.

Each value has an associated name and type (also known as a data type). The name is used to access the object. This name can be a simple identifier, or it can be a complex expression that uniquely references the object.

Lvalues

An lvalue is an object locator: an expression that designates an object. An example of an lvalue expression is `*P`, where `P` is any expression evaluating to a non-null pointer. A modifiable lvalue is an identifier or expression that relates to an object that can be accessed and legally changed in memory. A constant pointer to a constant, for example, is not a modifiable lvalue. A pointer to a constant can be changed (but its dereferenced value cannot).

Rvalues

The expression `a + b` is not an lvalue: `a + b = a` is illegal because the expression on the left is not related to an object. Such expressions are sometimes called rvalues (short for right values).

4.1.3 Scope and visibility

Scope

The scope of identifier is the part of the program in which the identifier can be used to access its object. There are different categories of scope: block (or local), function, function prototype, and file. These depend on how and where identifiers are declared.

Block: The scope of an identifier with block (or local) scope starts at the declaration point and ends at the end of the block containing the declaration (such a block is known as the enclosing block). Parameter declarations with a function definition also have block scope, limited to the scope of the function body.

File: File scope identifiers, also known as globals, are declared outside of all blocks; their scope is from the point of declaration to the end of the source file.

Function: The only identifiers having function scope are statement labels.